Paper 13 - Standard-Model Quark-Face Transport

CQE-paper-13
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Construction Basis

Map color-state and interaction analogs into a disciplined quark-face transport read, without overclaiming physical proof.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-13.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-13\paper_kernel_manifest.json
- body: production\papers\CQE-paper-13\PAPER-BODY.md
- source: production\papers\CQE-paper-13\SOURCE.md
- formal: production\papers\CQE-paper-13\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-13\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-13\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-13.md

Paper 13 - Standard-Model Quark-Face Transport

Abstract

This paper promotes the Standard-Model quark-face layer only to the extent currently supported by finite algebraic receipts. The closed result is not a derivation of physical quarks. It is the disciplined transport statement beneath that proposed reading: the shell-2 `LCR` chart has exactly three states, those states correspond to the three trace-2 idempotents of `J_3(O)`, the six-element `S_3` Weyl action closes on that triple, and the `n=3` shell-2 transition is an exact element of the `SU(3)` Weyl group ring over the rationals. A bounded `G2/F4/T5A` route classifies already-enumerated bits in at most three stages while explicitly refusing to become a cold-start depth-only derivation.

The six-face color/anticolor workbook model is retained as an analog exposure surface. It is useful because it makes the algebraic transport inspectable, but it does not by itself prove physical color charge, CKM phase, weak parity, or a complete Standard Model extension.

Claims

Claim 13.1. The shell-2 chart stratum is the three-element set `{(1,1,0), (1,0,1), (0,1,1)}`.

Claim 13.2. The three shell-2 states map bijectively to the three trace-2 idempotents `{E11+E22, E11+E33, E22+E33}` in `J_3(O)`.

Claim 13.3. The six permutations of diagonal indices in `S_3` close on the trace-2 triple.

Claim 13.4. The `n=3` shell-2 transition is an exact `SU(3)` Weyl group-ring element with residual squared equal to `0` over `Q`.

Claim 13.5. The bounded `G2/F4/T5A` route is a route classifier for an already-enumerated bit. It is not a depth-only derivation of that bit.

Claim 13.6. The color/anticolor six-face model is admitted as an analog transport model, not as a physical derivation.

Definitions

An **LCR chart state** is a triple `(L,C,R)` in `{0,1}^3`.

The **shell** of a chart state is `L + C + R`.

The **shell-2 stratum** is the set of chart states with shell value `2`.

A **quark face** in this paper is an algebraic face: one member of the trace-2 idempotent triple of `J_3(O)`. The word "quark" names the intended Standard Model analogy, not a completed physical identification.

The **color Weyl action** is the `S_3` action induced by permuting diagonal indices `(1,2,3)` and then reading the induced permutation of trace-2 idempotent pairs.

A **bounded route classifier** is a route that may classify an already-supplied enumeration value while preserving a visible boundary that it did not derive the value from depth alone.

Theorem 13

The CQECMPLX quark-face layer is a closed algebraic transport layer:

```
shell-2 LCR triple
-> trace-2 J_3(O) idempotent triple
-> closed S_3 Weyl action
-> exact n=3 SU(3) group-ring closure
-> bounded exceptional route classification
```

and the physical Standard Model reading remains an explicit obligation unless and until a separate calibrated derivation is supplied.

Proof

First enumerate all binary chart states with shell value `2`. There are exactly three:

```
C- = (1,1,0)
C0 = (1,0,1)
C+ = (0,1,1)
```

This proves Claim 13.1 by exhaustion.

Next map these states to the trace-2 idempotents:

```
C- -> E11 + E22
C0 -> E11 + E33
C+ -> E22 + E33
```

`verify_j3o_axioms` verifies that the diagonal idempotents are idempotent and Jordan-orthogonal, that they sum to the identity, and that the three trace-2 objects have trace `2` and are idempotent. This proves Claim 13.2 at the algebraic layer.

Now let a permutation `sigma` in `S_3` act on diagonal indices. For any trace-2 pair `{i,j}`, the image is `{sigma(i), sigma(j)}`, again one of the three two-element diagonal pairs. Since all six permutations are enumerated and every image lands inside the same three-element set, the Weyl action closes on the quark-face triple. This proves Claim 13.3.

The exact transition check is stronger than a floating-point fit. The verifier `verify_n3_su3_closure_exact` computes the `n=3` shell-2 conditional matrix and decomposes it over the `S_3` permutation matrices using rational arithmetic. The receipt reports residual squared equal to `0` and `is_exact_group_ring_element = true`. This proves Claim 13.4.

The exceptional route layer is then admitted with its honesty boundary intact. `verify_conjugate_triple(max_depth=256)` reports a passing bounded classifier: the route is oracle-backed, all tested routes resolve in at most three stages, and `depth_only_bridge` is false. Therefore it classifies supplied bits but does not claim to derive them cold-start from depth. This proves Claim 13.5 while also limiting it.

Finally, the six-face model has three color faces, three conjugate faces, involutive conjugation, and a closed `Z3` cycle on the three color faces. That is a valid analog model of the algebraic transport surface. It is not a derivation of physical color charge. This proves Claim 13.6 with the required boundary.

Combining the five closed algebraic layers and the explicit physical boundary proves the theorem.

Receipt

The promoted verifier is:

```
production/formal-papers/CQE-paper-13/verify_quark_face_transport.py
```

It writes:

```
production/formal-papers/CQE-paper-13/quark_face_transport_receipt.json
```

The closed layers are:

```
three shell-2 chart states
three trace-2 J_3(O) idempotents
six S3 Weyl actions close on the trace-2 triple
n=3 shell-2 transition is exact over the SU(3) Weyl group ring
bounded G2/F4/T5A route classifies oracle-enumerated bits in <=3 stages
six-face color/anticolor analog model is internally consistent
```

The open layers are:

```
derivation of physical quark color charge
derivation of measured CKM phase or V-A weak structure
cold-start depth-only derivation through the G2/F4/T5A route
```

Falsifiers

This paper fails if the shell-2 stratum does not contain exactly three states.

It fails if any trace-2 idempotent check in `J_3(O)` fails.

It fails if any `S_3` action leaves the trace-2 triple.

It fails if exact `n=3` closure has nonzero residual.

It fails if the bounded route is presented as a cold-start derivation.

It fails if the analog color-face model is used as a physical Standard Model proof without a separate calibrated derivation.

Role in the Suite

Paper 12 supplies a local CA prediction surface. Paper 13 receives one of its correction/transport fields and shows how it can be read through a three-face `J_3(O)` and `SU(3)`-Weyl closure. Paper 14 may use this as boundary-curvature input only if it cites the receipt and preserves the physical-obligation boundary.

Conclusion

Paper 13 closes a real algebraic layer: shell-2 chart states, trace-2 idempotents, `S_3` transport, exact rational `SU(3)` closure, and bounded exceptional-route classification. It does not yet close the physical Standard Model derivation. That separation is the point of the paper: the transport surface is proved; the physics identification remains a named target rather than an implied result.